

QAISUMAH, Saudi Arabia

WMO: 403730

Lat: 28.335N

Lon: 46.125E

Elev: 358

StdP: 97.10

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.5	(c) 5.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.9	(k) 13.8	(l) 9.6	(m) 14.1	(n) 1.7	(o) 270	(p) 0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.7	(c) 46.9	(d) N/A	(e) 45.9	(f) N/A	(g) 44.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.2	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 8.9	(c) 7.7	(e) 0.3	(f) 48.6	(g) 2.1	(h) 1.5	(i) -1.2	(j) 49.6	(k) -2.4	(l) 50.5	(m) -3.6	(n) 51.3	(o) -5.1	(p) 52.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	12.6	15.1	20.0	25.9	32.2	35.5	37.0	37.2	33.9	28.6	19.5	14.0		
	DBStd	9.47	3.24	3.56	3.76	3.59	2.84	1.97	1.94	1.92	2.26	3.01	3.90	3.39		
	HDD10.0	20	11	3	0	0	0	0	0	0	0	0	0	6		
	HDD18.3	474	179	99	24	1	0	0	0	0	0	0	32	139		
	CDD10.0	5872	92	148	310	478	689	766	838	843	717	576	286	129		
	CDD18.3	3284	2	10	76	229	431	516	580	585	467	318	68	4		
	CDH23.3	53418	26	106	794	2788	6897	9182	10599	10591	7698	4086	601	50		
	CDH26.7	37549	3	23	312	1488	4656	6799	8121	8115	5404	2408	214	6		
(14) Wind	WSAvg	3.4	3.1	3.5	3.7	3.9	3.7	3.8	3.6	3.2	2.8	3.1	3.0	3.1		
(15) Precipitation	PrecAvg	97	17	15	8	14	2	0	0	0	0	4	19	19		
	PrecMax	230	53	50	52	53	18	4	1	0	0	40	88	97		
	PrecMin	26	0	3	0	1	0	0	0	0	0	0	0	0		
	PrecStd	54	16	15	15	13	4	1	0	0	0	10	27	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	29.9	36.3	40.2	44.7	46.9	48.1	48.0	45.8	41.1	34.1	27.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	23.1	26.2	32.9	38.0	42.8	45.4	46.9	47.0	44.2	39.2	31.2	24.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	20.4	24.0	30.0	36.0	41.2	44.2	45.8	45.9	42.9	37.9	29.2	22.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	18.8	22.0	27.5	34.0	40.0	43.0	44.4	44.8	41.5	36.2	27.0	20.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	12.0	13.0	13.6	14.1	14.8	14.7	15.1	15.1	14.4	11.7	10.9		
		MCDBR	13.4	14.4	16.0	15.6	15.4	15.7	15.4	15.5	15.8	15.4	14.5	13.5		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.464	0.516	0.580	0.668	0.690	0.626	0.610	0.589	0.551	0.544	0.523	0.472		
	taud		1.833	1.717	1.589	1.473	1.447	1.530	1.579	1.620	1.678	1.692	1.732	1.824		
	Ebn at Noon		742	740	723	678	666	706	716	727	737	704	676	708		
	Edh at Noon		178	218	263	304	312	286	272	259	237	220	196	173		
(44) All-Sky Solar Radiation	RadAvg		3.76	4.65	5.68	6.23	7.04	7.90	7.63	7.18	6.42	5.18	3.83	3.43		
	RadStd		0.25	0.26	0.24	0.26	0.34	0.26	0.20	0.15	0.18	0.23	0.31	0.28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	+207	+170
(47) Regional (4 neighbors)		+0.54	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	+207	+170

Nomenclature: See separate page